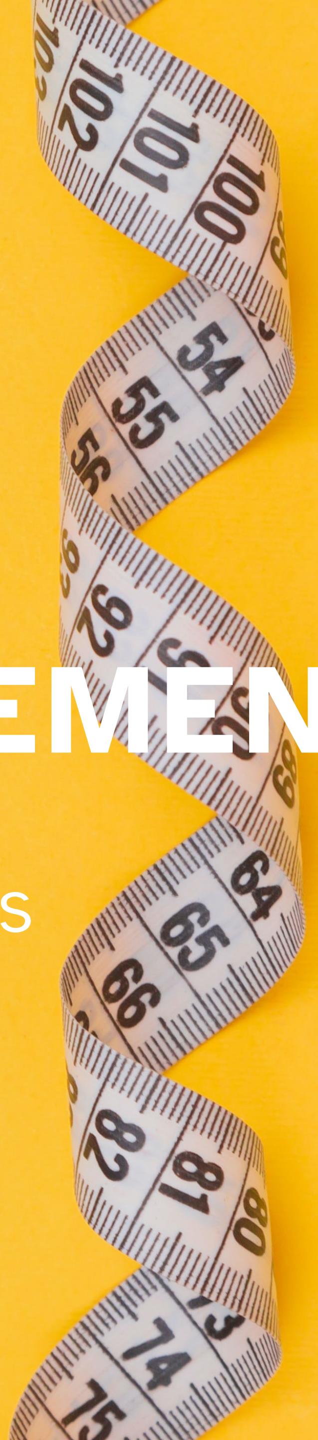
1
SCIENCE & MEASUREMENT

UNIT 01: FOUNDATIONAL CHEMISTRY SKILLS



CHEMISTRY 1310

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LEARNING OBJECTIVES

CHAPTER 1

1. Classify matter
2. Distinguish between physical and chemical changes and properties
3. Outline the principles of the scientific approach to molecules
4. Report scientific measurements to reflect certainty
5. Work with significant figures
6. Use conversion factors

The content and learning objectives addressed in Chapter 1 will carry throughout the entirety of the course, in both the lecture and laboratory components.

You are expected to review the material on your own, and may find that much of it is review.

Classification of Matter

1.1

Matter is anything that has mass and occupies space.

- Matter is classified according to its state (solid, liquid, or gas) and according to its composition (the kinds and amounts of substances that compose it).

Pure substances are those composed of *only* either a single type of atom or a single type of molecule.

- Elements are substances that cannot be chemically broken down into simpler substances.
- Compounds are composed of two or more elements in fixed, definite proportions.

Mixtures are composed of two or more different types of atoms or molecules that can be combined in variable proportions.

- Heterogeneous mixtures have compositions that vary from one region to another; e.g. wet sand
- Homogeneous mixtures have the same composition throughout; e.g., sugar water

Classification of Matter

1.1

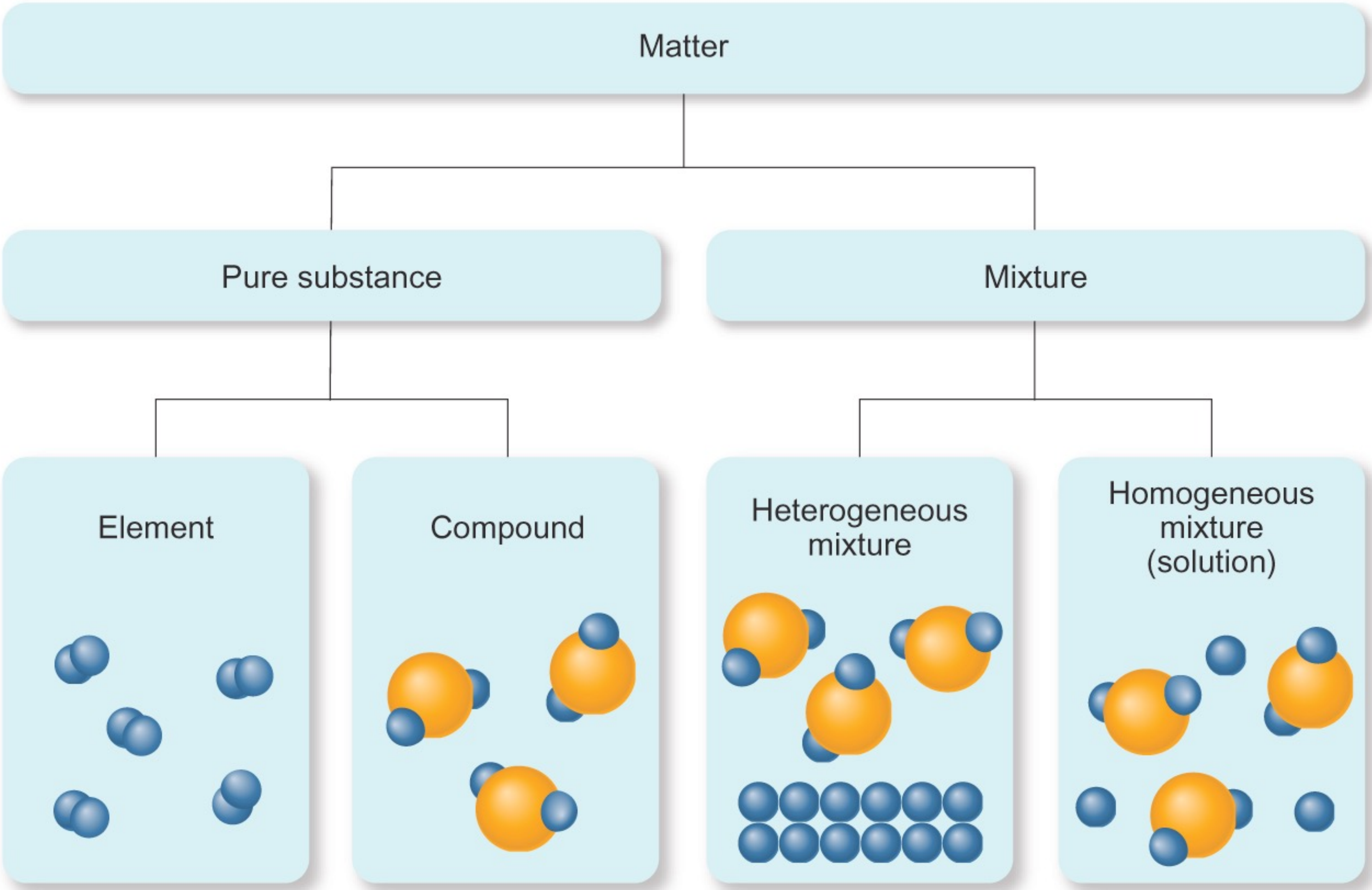


FIGURE 1.2

Properties of Matter

1.2

Properties are characteristics by which matter or substances can be identified.

- Examples: color, odor, state of matter at room temperature, etc.

Physical properties are properties that substances display without changing their compositions (odor, taste, color, melting point, density, etc.). Physical changes alter only the state or appearance but not the composition.

Chemical properties are those that substances display only by changing composition via chemical change (corrosiveness, flammability, acidity, toxicity). Chemical changes alter the composition of matter through the rearrangement of atoms to transform the original substance(s) into different substance(s).

Extensive properties are properties that depend on the quantity of the sample (mass).

Intensive properties are properties that remain unchanged regardless of sample size (boiling point).

The Scientific Method, Hypotheses, Theories, and Laws

1.4

Hypothesis

- A tentative interpretation or explanation of observations
- Should be falsifiable—it makes predictions that can be supported or refuted by further observation

Experiments

- Highly controlled procedures designed to generate observations that can support or refute a hypothesis

The Scientific Method, Hypotheses, Theories, and Laws

1.4

Scientific law

- A brief **statement** that summarizes past **observations** and predict future ones
- Example: The law of conservation of mass states, “In a chemical reaction, matter is neither created nor destroyed.”
- Laws are like hypotheses in that they are subject to experiments which can add support to them or prove them wrong.

Scientific theory

- A **model** for the way nature is that attempts to **explain** not merely what nature does, but why.
- Often, theories *predict* behavior far beyond the observations or laws from which they were developed.
- Example: Dalton’s atomic theory proposed that matter is composed of small, indestructible particles (atoms) that rearrange during chemical changes such that the total amount of mass remains constant.

The Scientific Method, Hypotheses, Theories, and Laws

1.4

Laws are scientific observations that have always been seen to be true, but they are not explanations for *why* the observation is always that way.

Theories are scientific explanations that so far have never been proven wrong.

- A theory and a law are two entirely different things.
- A hypothesis can become a theory, but a theory can never become a law.

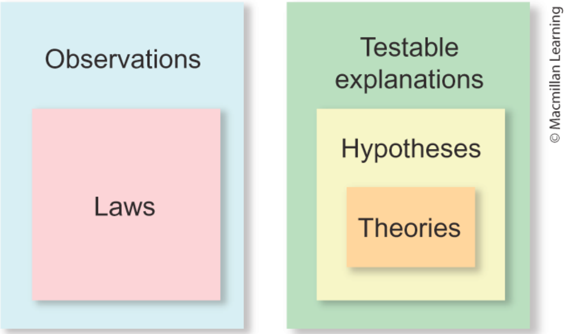


FIGURE 1.9

Scientific knowledge is empirical

- It is based on observation and experiment.

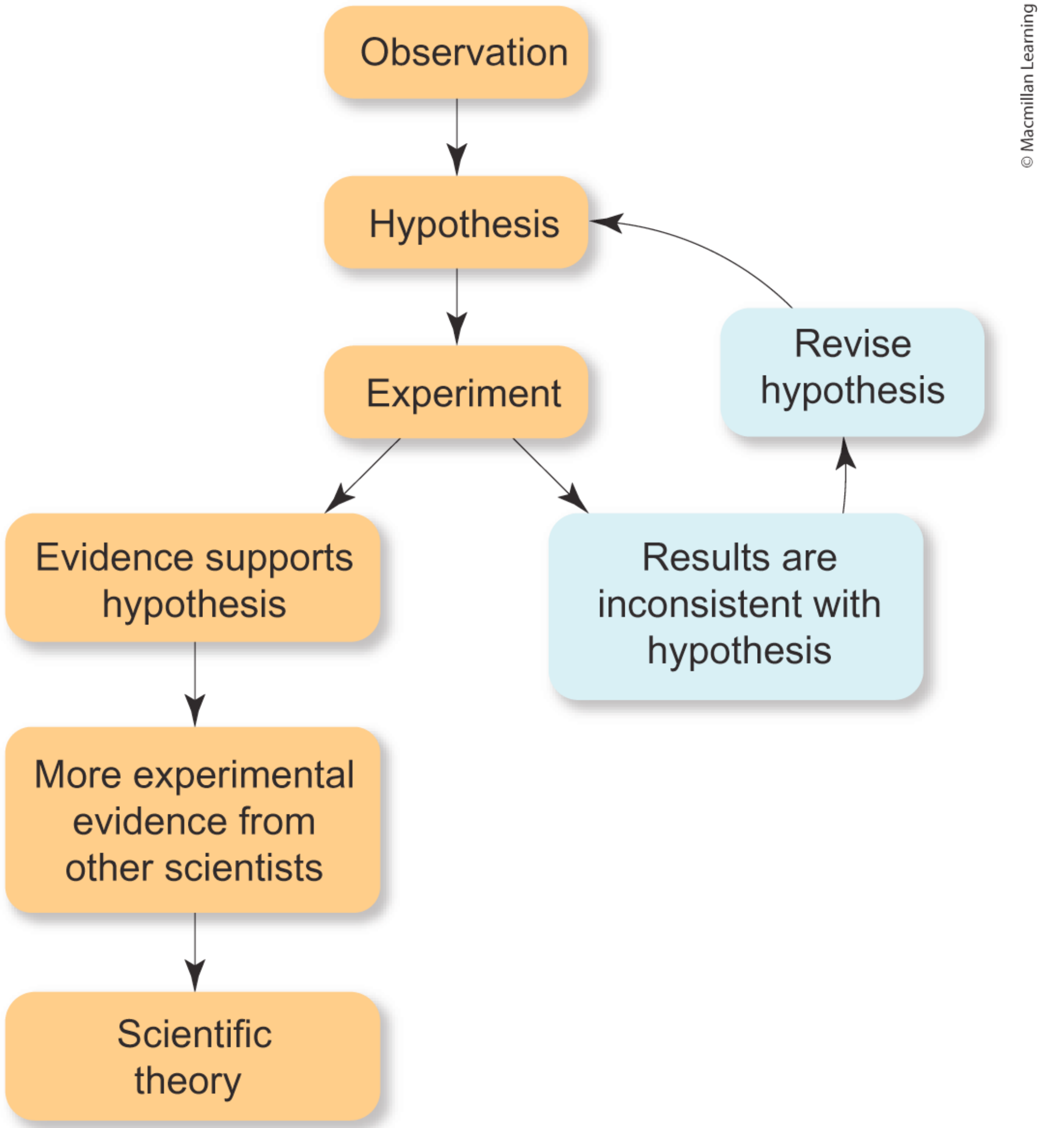


FIGURE 1.8

The International System of (SI) Units

1.5

Scientists use the International System of Units (SI), which is based on the metric system.

You should be familiar with the following seven base units.

Measurement	Unit	Symbol
Length	Meters	m
Mass	Kilograms	kg
Time	Seconds	s
Temperature	Kelvin	K
Amount of Substance	Moles	mol
Electric Current	Amperes	A
Luminous Intensity	Candela	cd

- *Note: We will likely not use luminous intensity in this course.*

The International System of (SI) Units

1.5

Scientific notation allows us to express very large or very small quantities in a compact way by using negative and positive exponents.

- The SI system uses prefix multipliers along with the standard units.
- You should become familiar with the following prefixes.

All other measurements are measured in units that *derived* from some combinations of the seven base units. For example,

- Speed: m/s or km/hr
- Volume: 1 cm^3, 1 mL, 10^{-3} L
- Density: mass / volume, g/cm^3

Prefix	Abbreviation	Numerical Factor
Tera-	T	$\times 10^{12}$
Giga-	G	$\times 10^{9}$
Mega-	M	$\times 10^{6}$
Kilo-	k	$\times 10^{3}$
Deci-	d	$\times 10^{-1}$
Centi-	c	$\times 10^{-2}$
Milli-	m	$\times 10^{-3}$
Micro-	µ	$\times 10^{-6}$
Nano-	n	$\times 10^{-9}$
Pico-	p	$\times 10^{-12}$
Femto-	f	$\times 10^{-15}$
Atto-	a	$\times 10^{-18}$

Significant Digits

1.6

Accuracy refers to how close the measured value is to the actual value.

Precision refers to how close a series of measurements are to one another or how reproducible they are.

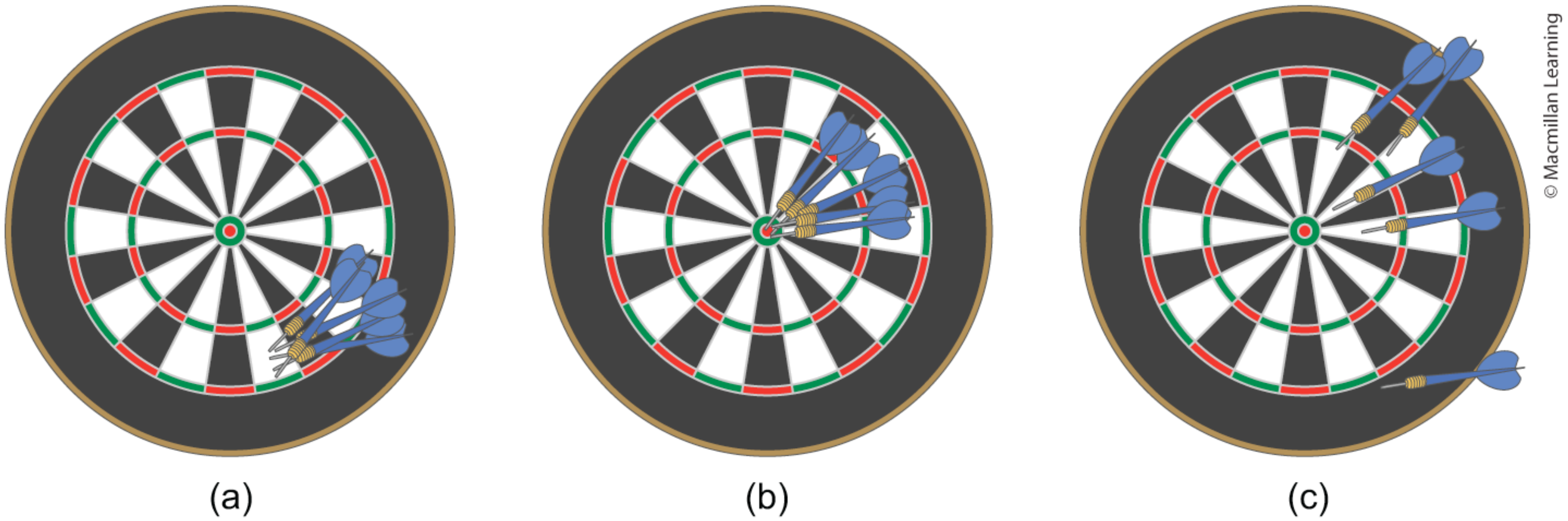


FIGURE 1.13 **Precise but not accurate** **Both precise and accurate** **Neither precise nor accurate**

Significant Digits

1.6

Scientific measurements are reported so that every digit is certain *except* the last, which is estimated.

- The number of digits reported in a measurement depends on the measuring device.
- Always estimate to one digit beyond the smallest scale division, if possible.

Significant figures or significant digits are all the certain digits *and* the last uncertain (estimated) digit.

- This is a *simple* system to represent errors in measurements by localizing all the uncertainty into *only* the final digit.
- The greater the number of significant figures, the greater the certainty of the measurement.
- For example: 1.01 cm vs. 1.001 vs. 1.0001 cm

Significant Digits

1.6

Rules for Determining Significant Figures

1. All nonzero digits are significant
2. Zeroes between two significant figures are themselves significant
3. Zeroes at the beginning of a number are never significant
4. Zeroes at the end of a number are significant and after the decimal point are always significant
5. Zeros at the end of a number and before the decimal point are not significant if a decimal point is not present.
    - Write these numbers in scientific notation to clarify if a zero is significant.

Exact numbers have no uncertainty, and thus do not limit the number of significant figures in any calculations.

They originate from three sources:

- Accurate counting of discrete objects
- Defined quantities or defined constants
- Integral numbers that are part of an equation

Significant Digits

1.6

Calculations: Dealing with significant figures when adding or subtracting measurements

- The answer has the same number of decimal places as there are in the measurement with the fewest decimal places.
- In other words, the sum or different (calculated answer) is reported to the same precision of the less precise measurement.

Example: adding two volumes

$$\begin{array}{r} 83.\underline{5}\ \ \ \text{mL} \\ \underline{+\ 23.28\ \text{mL}} \\ 106.\underline{7}8\ \text{mL} \end{array} = 106.8\ \text{mL}$$

Example: subtracting two volumes

$$\begin{array}{r} 865.\underline{9}\ \ \ \ \ \ \ \text{mL} \\ \underline{-\ \ \ 2.8121\ \text{mL}} \\ 863.\underline{0}879\ \text{mL} \end{array} = 863.1\ \text{mL}$$

Significant Digits

1.6

Calculations: Dealing with significant figures when multiplying or dividing measurements

- The number with the least certainty limits the certainty of the result.
- Therefore, the answer contains the same total number of significant figures as there are in the measurement with the fewest significant figures.

- Example: multiplying the following measurements to find volume

$$\text{Volume} = \underbrace{9.2}_{\text{2 significant figures}}\ \text{cm} \times 6.85\ \text{cm} \times 0.3744\ \text{cm} = 23.5947\ \text{cm}^3 = \underbrace{24}_{\text{2 significant figures}}\ \text{cm}^3$$

Significant Digits

1.6

Calculations: How to round with significant figures

- There are different styles of rounding. We will use a simplified approach.

1. If the digit removed is 5 or greater, then the preceding number increases by 1.
   - Example: 5.379 rounds to 5.38 if three significant figures are retained.
   - Example: 5.379 rounds to 5.4 if two significant figures are retained.
2. If the digit removed is less than 5, the preceding number is unchanged.
   - Example: 0.2413 rounds to 0.241 if three significant figures are retained
   - Example: 0.2413 rounds to 0.24 if two significant figures are retained.
3. Be sure to carry *at least one additional significant figure* throughout a multistep calculation.
4. **Round off only at the final answer.**

Dimensional Analysis

1.7

We use dimensional analysis to convert one quantity to another.

Most commonly, dimensional analysis utilizes conversion factors.

- For example, the conversion factor between inches and centimeters is 1 in = 2.54 cm
- Note that the conversion factor can be written as a fraction in either form below depending on how it is used.

$$\frac{1\ \text{in}}{2.54\ \text{cm}} \quad \text{or} \quad \frac{2.54\ \text{cm}}{1\ \text{in}}$$

- Use the form of the conversion factor that puts the desired unit in the numerator.

$$\cancel{\text{Given}}\ \text{unit} \times \frac{\text{desired unit}}{\cancel{\text{given}}\ \text{unit}} = \text{desired unit}$$

- Example: Convert 8.00 m to inches

$$8.00\ \text{m} \times \frac{100\ \text{cm}}{1\ \text{m}} \times \frac{1\ \text{in}}{2.54\ \text{cm}} = 315\ \text{in}$$

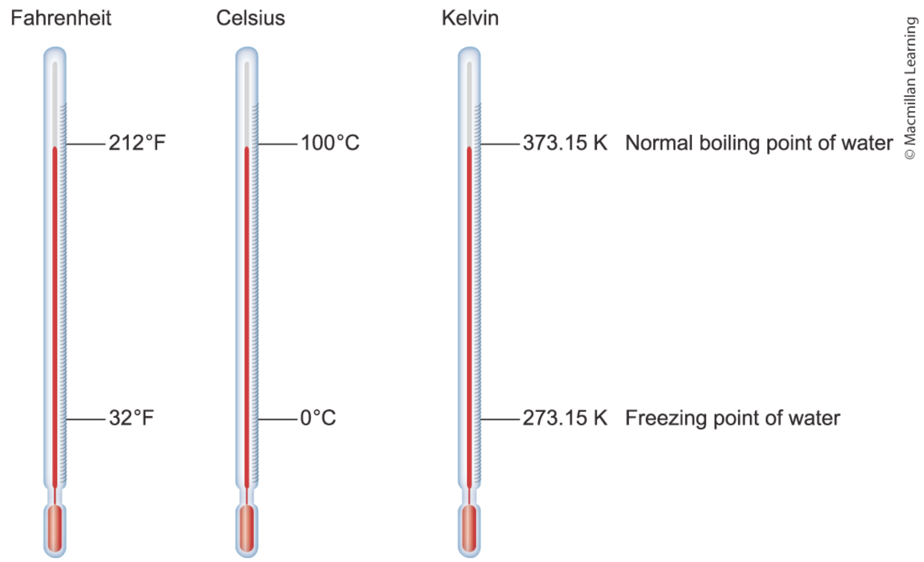
Temperature Scales

1.9

In scientific measurements Celsius and Kelvin scales are most often used.

- The Celsius scale is based on the properties of water.
- 0 °C is the freezing point of water.
- 100 °C is the boiling point of water.

FIGURE 1.21 Comparing the Temperature Scales



Kelvin is the SI unit of temperature.

- Based on the properties of gases.
- There are no negative Kelvin temperature.
- To convert from a temperature in degrees Celsius (°C) to the temperature in Kelvin (K) add 273.15

$$T_{\mathrm{K}} = T_{^\circ\mathrm{C}} + 273.15$$

The Fahrenheit scale is not used in scientific measurements.